

THE LOWEST-WEIGHT COHOMOLOGY OF $\mathcal{M}_{6,6}$ IS TATE

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ABSTRACT. We determine the Hodge type of the lowest-weight part of $H^{26}(\mathcal{M}_{6,6}; \mathbf{Q})$. We prove that

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) = \mathrm{RH}^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-13)^{\oplus r}$$

for some $r \geq 0$. Equivalently, $W_{-16}H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus r}$. The proof combines the description by Canning–Larson–Payne of the primitive quotient of the lowest-weight cohomology of $\mathcal{M}_{g,n}$ with Harer’s computation of the virtual cohomological dimension. The primitive quotient vanishes in the case at hand, as does the contribution pulled back from $\mathcal{M}_{6,5}$. The remaining classes are products of cotangent-line classes with lowest-weight classes on $\mathcal{M}_{6,5}$; the latter are algebraic by the sharp genus-six Chow–Künneth generation result. The argument does not determine the multiplicity r .

1. INTRODUCTION

Let $\mathcal{M}_{g,n}$ be the Deligne–Mumford stack of smooth complex curves of genus g with n ordered marked points. We work throughout with rational cohomology and with the mixed Hodge structures carried by the cohomology of complex algebraic stacks. The group considered in this paper is

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}),$$

the lowest-weight part of the degree-26 cohomology. Its Hodge type is the open contribution associated with the pair $(g, n) = (6, 6)$ in the degree-16 boundary induction for the moduli stack of stable curves. Indeed, $\dim_{\mathbf{C}} \mathcal{M}_{6,6} = 21$, and Poincaré duality identifies it, after a Tate twist, with the weight- -16 part of degree-16 Borel–Moore homology.

Our main result is the following.

Theorem 1.1. *There is a nonnegative integer r such that*

$$\begin{aligned} W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) &= \mathrm{RH}^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-13)^{\oplus r}, \\ W_{-16}H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) &\cong \mathbf{Q}(8)^{\oplus r}, \\ \mathrm{Gr}_{16}^W H_c^{16}(\mathcal{M}_{6,6}; \mathbf{Q}) &\cong \mathbf{Q}(-8)^{\oplus r}. \end{aligned}$$

Here RH^* denotes tautological cohomology. The assertion includes the possibility $r = 0$.

The proof is short once two published structural results are put together at their exact numerical limits. Canning, Larson, and Payne describe the quotient of $W_k H^k(\mathcal{M}_{g,n})$ by classes pulled back from one fewer marking and by their products with cotangent-line classes [3, Lemma 3.1]. At $(g, n, k) = (6, 6, 26)$, the quotient is built from degree-20 cohomology of symplectic local systems on $\mathcal{M}_6$. Harer’s formula gives $\mathrm{vcd}(\mathrm{Mod}_6) = 19$, so every such group vanishes. The part pulled back from $\mathcal{M}_{6,5}$ vanishes as well, because $\mathrm{vcd}(\mathrm{PMod}_{6,5}) = 25$.

It remains to determine the Hodge type of classes obtained by multiplying pullbacks from $W_{24} H^{24}(\mathcal{M}_{6,5})$ by cotangent-line classes. This is where the sharp endpoint of the known Chow–Künneth generation range is essential: the genus-six bound is five marked points [3, Table 1]. Proposition 4.5 of [3] therefore shows that $W_{24} H^{24}(\mathcal{M}_{6,5})$ is tautological and hence of type $(12, 12)$. Multiplication by a cotangent-line class gives type $(13, 13)$.

The argument uses virtual cohomological dimension only in degrees strictly above the relevant bounds: degree $20 > 19$ on $\mathcal{M}_6$ and degree $26 > 25$ on $\mathcal{M}_{6,5}$. In particular, it does not infer any vanishing from the equality $26 = \mathrm{vcd}(\mathrm{PMod}_{6,6})$. We also stress that Theorem 1.1 determines the Hodge type, but not the dimension, of the group.

2. CONVENTIONS AND THE FORGETFUL SUBSPACES

The Tate Hodge structure $\mathbf{Q}(m)$ has weight $-2m$. For a smooth complex algebraic stack X , the weights on $H^j(X; \mathbf{Q})$ are at least j ; consequently $W_j H^j(X; \mathbf{Q})$ is a pure Hodge structure of weight j . We use the increasing weight filtration. Our conventions agree with those of [3]. The basic mixed Hodge theory is due to Deligne [4]; rational cohomology and Poincaré duality for Deligne–Mumford stacks may be obtained from a finite level cover and are also discussed in [1].

For $1 \leq i \leq n$, let

$$\pi_i: \mathcal{M}_{g,n} \longrightarrow \mathcal{M}_{g,n-1}$$

be the map forgetting the i th marking, and let $\psi_i = c_1(L_i) \in H^2(\mathcal{M}_{g,n}; \mathbf{Q})$ be the first Chern class of the corresponding cotangent line. Following [3, Sections 2–3], define

$$\Phi_{g,n}^k := \sum_{i=1}^n \pi_i^* W_k H^k(\mathcal{M}_{g,n-1}; \mathbf{Q}), \quad (1)$$

$$\Psi_{g,n}^k := \sum_{i=1}^n \psi_i \smile \Phi_{g,n}^{k-2}. \quad (2)$$

Thus $\Phi_{g,n}^k$ consists of lowest-weight classes inherited from one fewer marking, while $\Psi_{g,n}^k$ is generated by such classes after one multiplication by a cotangent-line class.

Every algebraic cycle of codimension a has cohomology class of Hodge type (a, a) and weight $2a$. In particular,

$$\mathrm{RH}^k(\mathcal{M}_{g,n}; \mathbf{Q}) \subseteq W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}),$$

with both sides zero in odd degree for the tautological contribution. The classes ψ_i are tautological and have type $\mathbf{Q}(-1)$.

3. THE PRIMITIVE QUOTIENT AND ALGEBRAICITY AT FIVE MARKINGS

We recall the two results about pure cohomology that will be used in the proof. Let $\mathbb{V}_\lambda$ denote the irreducible symplectic local system on $\mathcal{M}_g$ indexed by a partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_g \geq 0)$, and put $|\lambda| = \sum_i \lambda_i$. If $|\lambda| = n$, write $V_{\lambda^\top}$ for the corresponding rational representation of the symmetric group; as a Hodge structure it has type $(0, 0)$.

Proposition 3.1 (Canning–Larson–Payne). *If $g \geq 2$, then there is an isomorphism of pure rational Hodge structures*

$$\frac{W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})}{\Phi_{g,n}^k + \Psi_{g,n}^k} \cong \bigoplus_{|\lambda|=n} W_k H^{k-n}(\mathcal{M}_g; \mathbb{V}_\lambda) \otimes V_{\lambda^\top}. \quad (3)$$

Proof. This is [3, Lemma 3.1(a)]. We briefly recall the geometry behind the formula. If $\pi: \mathcal{C} \rightarrow \mathcal{M}_g$ is the universal closed curve, then $\mathcal{M}_{g,n}$ is the complement of the pairwise diagonals in the n -fold fiber power $\mathcal{C}^n$ over $\mathcal{M}_g$. Restriction from the fiber power surjects onto the lowest-weight cohomology of $\mathcal{M}_{g,n}$. The smooth-proper Leray decomposition for $\mathcal{C}^n \rightarrow \mathcal{M}_g$ has Künneth indices in $\{0, 1, 2\}^n$. The summands with a zero index are generated by forgetful pullbacks. More precisely, the identification of the index-two summands modulo the zero-index summands is made by the relative Chow–Künneth projector π_2 of Petersen–Tavakol–Yin [6, Sections 5.1 and 5.2.2]; it is not inferred merely from the restriction of a cotangent-line class to a fiber. The image of this projector is generated by the cotangent-line contribution in (2). The all-index-one term is the n th tensor power of $R^1 \pi_* \mathbf{Q}$; quotienting by contractions coming from the diagonals leaves precisely the symplectic local systems of weight n in (3). $\square$

The second input identifies lowest-weight cohomology with algebraic classes at the endpoint of the presently known genus-six range.

Proposition 3.2. *For every k ,*

$$W_k H^k(\mathcal{M}_{6,5}; \mathbf{Q}) = \mathrm{RH}^k(\mathcal{M}_{6,5}; \mathbf{Q}).$$

In particular, for some $a \geq 0$,

$$W_{24} H^{24}(\mathcal{M}_{6,5}; \mathbf{Q}) \cong \mathbf{Q}(-12)^{\oplus a}.$$

Proof. Table 1 of Canning–Larson–Payne records the inclusive genus-six bound $c(6) = 5$ for the Chow–Künneth generation property and tautological

Chow ring of the open stack $\mathcal{M}_{6,n}$. Their Lemma 4.3 and Proposition 4.5 then give [3, Table 1, Lemma 4.3, and Proposition 4.5]

$$W_k H^k(\mathcal{M}_{6,5}; \mathbf{Q}) = \mathrm{RH}^k(\mathcal{M}_{6,5}; \mathbf{Q})$$

for every k . In degree 24, every class on the right is the class of a rational linear combination of codimension-12 tautological cycles, so the Hodge structure is concentrated in type $(12, 12)$. It is therefore a finite direct sum of $\mathbf{Q}(-12)$. $\square$

4. VIRTUAL COHOMOLOGICAL DIMENSION

Let Mod_g be the mapping class group of a closed oriented surface of genus g , and let $\mathrm{PMod}_{g,n}$ be the pure mapping class group with n ordered punctures. The analytic stacks $\mathcal{M}_g$ and $\mathcal{M}_{g,n}$ have the homotopy types of the corresponding classifying stacks, since Teichmüller space is contractible. Hence cohomology of a rational local system agrees with group cohomology of the associated rational module.

Lemma 4.1. *Let Γ be a virtually torsion-free group of virtual cohomological dimension d . If L is a rational Γ -module, then*

$$H^q(\Gamma; L) = 0 \quad (q > d).$$

Proof. Choose a torsion-free normal subgroup $\Gamma' \triangleleft \Gamma$ of finite index with cohomological dimension d . Then $H^q(\Gamma'; L) = 0$ for $q > d$. In the Hochschild–Serre spectral sequence for $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow F \rightarrow 1$, the group F is finite. Since the coefficient field has characteristic zero, taking F -invariants is exact and $H^p(F; U) = 0$ for $p > 0$ and every rational F -module U . The spectral sequence therefore gives

$$H^q(\Gamma; L) \cong H^q(\Gamma'; L)^F,$$

which vanishes for $q > d$. See also [2, Chapter VII]. $\square$

Harer computed [5, Theorem 4.1]

$$\mathrm{vcd}(\mathrm{Mod}_g) = 4g - 5, \quad \mathrm{vcd}(\mathrm{PMod}_{g,n}) = 4g - 4 + n \quad (n > 0), \quad (4)$$

for $g \geq 2$. The two instances needed below are therefore

$$\mathrm{vcd}(\mathrm{Mod}_6) = 19, \quad \mathrm{vcd}(\mathrm{PMod}_{6,5}) = 25.$$

Corollary 4.2. *The following vanishings hold:*

- (i) $H^{20}(\mathcal{M}_6; \mathbb{V}) = 0$ for every finite-dimensional rational local system $\mathbb{V}$;
- (ii) $H^{26}(\mathcal{M}_{6,5}; \mathbf{Q}) = 0$.

Proof. Apply Lemma 4.1 and (4). The first degree is $20 > 19$, and the second is $26 > 25$. $\square$

5. PROOF OF THE MAIN THEOREM

We first determine the entire lowest-weight group in ordinary cohomology.

Proposition 5.1. *One has*

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) = \Psi_{6,6}^{26}.$$

Proof. Apply Proposition 3.1 with $(g, n, k) = (6, 6, 26)$. Its right-hand side is

$$\bigoplus_{|\lambda|=6} W_{26}H^{20}(\mathcal{M}_6; \mathbb{V}_\lambda) \otimes V_{\lambda^\vee},$$

which vanishes by Corollary 4.2(i). It follows that

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) = \Phi_{6,6}^{26} + \Psi_{6,6}^{26}. \quad (5)$$

On the other hand, Corollary 4.2(ii) and the definition (1) give

$$\Phi_{6,6}^{26} = \sum_{i=1}^6 \pi_i^* W_{26}H^{26}(\mathcal{M}_{6,5}; \mathbf{Q}) = 0.$$

Substituting this in (5) proves the proposition. $\square$

Proof of Theorem 1.1. By Proposition 3.2, each summand used to generate $\Phi_{6,6}^{24}$ is a quotient of a finite direct sum of copies of $\mathbf{Q}(-12)$. Hence $\Phi_{6,6}^{24}$ is concentrated in Hodge type $(12, 12)$. Cup product with ψ_i , an algebraic class of type $(1, 1)$, is a morphism of Hodge structures

$$\mathbf{Q}(-1) \otimes \Phi_{6,6}^{24} \longrightarrow W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}).$$

It follows that $\Psi_{6,6}^{26}$ is concentrated in type $(13, 13)$. By Proposition 5.1, the same is true of the entire group $W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q})$. A rational Hodge structure concentrated in type $(13, 13)$ is, noncanonically, a direct sum of copies of $\mathbf{Q}(-13)$. Thus

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-13)^{\oplus r}$$

for some $r \geq 0$.

Moreover, Proposition 5.1 expresses every class in this group as a sum of products of tautological classes. Thus $W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) \subseteq \mathrm{RH}^{26}(\mathcal{M}_{6,6}; \mathbf{Q})$. The reverse inclusion holds because every codimension-13 tautological cycle class has weight 26. This proves the first line of the theorem.

It remains to translate the result into degree-16 homology. The complex dimension of $\mathcal{M}_{6,6}$ is

$$d = 3g - 3 + n = 21.$$

Poincaré duality for the smooth Deligne–Mumford stack $\mathcal{M}_{6,6}$ gives an isomorphism of mixed Hodge structures

$$H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong H^{2d-16}(\mathcal{M}_{6,6}; \mathbf{Q})(d) = H^{26}(\mathcal{M}_{6,6}; \mathbf{Q})(21).$$

Taking the indicated lowest-weight piece yields

$$W_{-16}H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q})(21) \cong \mathbf{Q}(-13)^{\oplus r}(21) = \mathbf{Q}(8)^{\oplus r}.$$

Finally, Borel–Moore homology and compactly supported cohomology are dual as mixed Hodge structures. With the standard dual weight filtration, this gives

$$\mathrm{Gr}_{16}^W H_c^{16}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathrm{Hom}_{\mathbf{Q}}\left(\mathrm{Gr}_{-16}^W H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}), \mathbf{Q}(0)\right).$$

The weights on $H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q})$ are at least -16 , so its weight- -16 graded piece is $W_{-16} H_{16}^{\mathrm{BM}}$. Since $\mathbf{Q}(8)^{\vee} = \mathbf{Q}(-8)$, the preceding Borel–Moore calculation therefore gives

$$\mathrm{Gr}_{16}^W H_c^{16}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus r},$$

as claimed. $\square$

6. REMARKS ON SCOPE

Remark 6.1 (What is proved). The proof determines the Hodge type of the full lowest-weight group, not only of a specified tautological subspace. In fact, it proves the stronger generation statement

$$W_{26} H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) = \Psi_{6,6}^{26} = \mathrm{RH}^{26}(\mathcal{M}_{6,6}; \mathbf{Q}).$$

Thus no non-Tate summand can survive in this group.

Remark 6.2 (What is not proved). The multiplicity r is not computed. Nor does the argument assert an integral result or identify the cohomology of the stack with that of its coarse moduli space. All statements are about rational mixed Hodge structures on Deligne–Mumford stacks over $\mathbf{C}$.

Remark 6.3 (Why the numerical endpoint matters). The only input that determines Hodge type, rather than merely generation, is Proposition 3.2 at $n = 5$. The bound $n \leq 5$ in genus six is inclusive and sharp for the present argument. No Chow–Künneth generation statement is applied directly to $\mathcal{M}_{6,6}$.

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